Class 9 Biology – The Fundamental Unit of Life (Annual Exam 2026)

Full Exam-Style Questions, Answers & Keywords (NCERT + CBSE Aligned)

Q1. Who discovered cells? Mention the year and the material observed.

Answer: Cells were discovered by **Robert Hooke** in the year **1665**. He observed a thin slice of **cork** under a primitive microscope and noticed small, box-like compartments, which he named *cells*.

Keywords: Robert Hooke, 1665, cork, microscope, box-like structures

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Q2. Why is the cell called the structural and functional unit of life?

Answer: The cell is called the structural and functional unit of life because all living organisms are made up of cells (structure) and all vital life processes such as respiration, digestion, excretion, growth, and reproduction occur within cells (function).

Keywords: basic unit, structure, function, life processes

Q3. Differentiate between prokaryotic and eukaryotic cells (any three points).

Answer: 1. Prokaryotic cells do not have a true nucleus, whereas eukaryotic cells have a well-defined nucleus. 2. Membrane-bound organelles are absent in prokaryotic cells but present in eukaryotic cells. 3. Prokaryotic cells are simpler and smaller, while eukaryotic cells are complex and larger.

Keywords: true nucleus, membrane-bound organelles, simple, complex

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Q4. State the cell theory. Who proposed it?

Answer: The cell theory states that: 1. All living organisms are made up of cells. 2. The cell is the basic unit of life. 3. All cells arise from pre-existing cells. The theory was proposed by **Schleiden and Schwann** and later modified by **Rudolf Virchow**.

Keywords: Schleiden, Schwann, Virchow, basic unit of life

Q5. Define diffusion and osmosis. State one difference between them.

Answer: Diffusion is the movement of particles from a region of higher concentration to lower concentration. Osmosis is the movement of water molecules through a selectively permeable membrane from a dilute solution to a concentrated solution.

Keywords: concentration gradient, water movement, selectively permeable membrane

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Q6. Why is the plasma membrane called selectively permeable?

Answer: The plasma membrane is called selectively permeable because it allows certain substances like oxygen, carbon dioxide, and water to pass through it while preventing the movement of harmful or unwanted substances.

Keywords: plasma membrane, selective entry, regulation

Q7. What is plasmolysis? Describe what happens to a plant cell in a hypertonic solution.

Answer: Plasmolysis is the process in which the cell membrane shrinks away from the cell wall when a plant cell is placed in a hypertonic solution due to loss of water by osmosis.

Keywords: hypertonic solution, water loss, shrinkage

Q8. Write the full forms of DNA and ATP.

Answer: DNA – Deoxyribonucleic Acid
ATP – Adenosine Triphosphate

Keywords: genetic material, energy currency

Q9. What is a nucleoid? Where is it found?

Answer: A nucleoid is an irregular region in a cell that contains genetic material but is not enclosed by a nuclear membrane. It is found in **prokaryotic organisms** such as bacteria.

Keywords: nucleoid, prokaryotes, no nuclear membrane

Q10. Explain how an Amoeba obtains its food.

Answer: Amoeba obtains food by the process of **endocytosis**. It extends finger-like projections called pseudopodia around the food particle and encloses it inside a food vacuole, where digestion occurs.

Keywords: pseudopodia, endocytosis, food vacuole

Q11. Why are lysosomes called the suicide bags of the cell?

Answer: Lysosomes contain powerful digestive enzymes that can break down worn-out organelles and foreign substances. If a cell is damaged, lysosomes may burst and digest the entire cell, causing its death.

Keywords: digestive enzymes, self-destruction

Q12. Which organelle is called the powerhouse of the cell? Why?

Answer: Mitochondria are called the powerhouse of the cell because they produce ATP during respiration. ATP provides energy required for all cellular activities.

Keywords: mitochondria, ATP, energy production

Q13. State three functions of the Golgi apparatus.

Answer: 1. Modifies proteins and lipids. 2. Packages materials into vesicles. 3. Helps in transport and secretion.

Keywords: modification, packaging, secretion

Q14. Differentiate between Rough ER and Smooth ER.

Answer: Rough ER has ribosomes and is involved in protein synthesis, whereas Smooth ER lacks ribosomes and is involved in lipid synthesis.

Keywords: ribosomes, proteins, lipids

Q15. Describe the structure and function of the nucleus.

Answer: The nucleus consists of a nuclear membrane, nucleolus, and chromatin material. It controls all cellular activities and stores genetic information, hence called the control center of the cell.

Keywords: control center, DNA, heredity

Q16. Why are mitochondria and plastids called semi-autonomous organelles?

Answer: Mitochondria and plastids have their own DNA and ribosomes, enabling them to synthesize some of their proteins independently.

Keywords: own DNA, ribosomes, semi-autonomous

Q17. Name the different types of plastids and their functions.

Answer: - Chloroplast: Photosynthesis - Chromoplast: Pigments and coloration - Leucoplast: Storage of food

Keywords: photosynthesis, pigments, storage

Q18. Why are vacuoles larger in plant cells than in animal cells?

Answer: Plant cells have a large central vacuole to maintain turgidity, store nutrients, and provide rigidity, whereas animal cells have small or no vacuoles.

Keywords: turgidity, storage, rigidity

Q19. Compare plant and animal cells (any three points).

Answer: Plant cells have a cell wall, plastids, and a large vacuole, whereas animal cells lack a cell wall, plastids, and have small vacuoles.

Keywords: cell wall, plastids, vacuole

Q20. What happens to an animal cell in hypotonic, isotonic, and hypertonic solutions?

Answer: - Hypotonic solution: Cell swells or bursts - Isotonic solution: Cell remains unchanged - Hypertonic solution: Cell shrinks

Keywords: osmosis, water movement, concentration difference

✔️Prepared strictly as per **NCERT & CBSE Class 9 (2026)** marking scheme